SI 140A-01 Probability & Statistics for EECS, Fall 2026
Homework 4

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Due on Oct. 27, 2026, 18:00 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1

A book has n typos. Two proofreaders, Prue and Frida, independently read the book. Prue catches each typo with probability p_1 and misses it with probability $q_1 = 1 - p_1$, independently, and likewise for Frida, who has probabilities p_2 of catching and $q_2 = 1 - p_2$ of missing each typo. Let X_1 be the number of typos caught by Prue, X_2 be the number caught by Frida, and X be the number caught by at least one of the two proofreaders.

(a) Find the distribution of X.

(b) For this part only, assume that $p_1 = p_2$. Find the conditional distribution of X_1 given that $X_1 + X_2 = t$.

(a) both of them miss the typo : $p = q_1 \cdot q_2 = (1-p_1)(1-p_2)$

$X \sim B(n, 1-q_1q_2)$, $P(X=k) = C_n^k (1-q_1q_2)^k \cdot (q_1q_2)^{n-k}$, $k = 0, 1, \cdots, n$

(b) $p_1 = p_2 = p$, X_1 and X_2 are independent,

$X_1 \sim B(n,p)$; $X_2 \sim B(n,p)$.

given $X_1 + X_2 = t$, $t \sim B(2n, p)$

$$P(X_1 = k \mid X_1 + X_2 = t) = \frac{P(X_1 = k \cap X_2 = t-k)}{P(X_1 + X_2 = t)} = \frac{P(X_1 = k) \cdot P(X_2 = t-k)}{P(X_1 + X_2 = t)} = \frac{C_n^k (1-q)^k (q^2)^{n-k} \cdot C_n^{t-k} (1-q)^{t-k} (q^2)^{n-t+k}}{C_{2n}^t (1-q)^t \cdot q^{2n-t}}$$

$$= \frac{C_n^k \cdot C_n^{t-k}}{C_{2n}^t}$$

$$P(X_1 = k \mid X_1 + X_2 = t) = \frac{C_n^k \cdot C_n^{t-k}}{C_{2n}^t} , \quad (0 \le k \le \min(n, t))$$

Problem 2

Let X be a random day of the week, coded so that Monday is 1, Tuesday is 2, *etc.* (so X takes values $1, 2, \ldots, 7$ with equal probabilities). Let Y be the next day after X. Do X and Y have the same distribution? What is $P(X < Y)$?

$P(X=k) = \frac{1}{7}$, $k = 1, 2, \cdots, 7$

$P(Y=k) = \frac{1}{7}$, $k = 1, 2, \cdots 17$

X and Y have the same distribution.

X	1	2	3	4	5	6	7
Y	2	3	4	5	6	7	1
	✓						✗

$P(X<Y) = \frac{6}{7}$

Problem 3

For X and Y binary digits (0 or 1), let $X \oplus Y$ be 0 if $X = Y$ and 1 if $X \neq Y$ (this operation is called exclusive or (often abbreviated to XOR), or addition mod 2).

(a) Let $X \sim \text{Bern}(p)$ and $Y \sim \text{Bern}(1/2)$, independently. What is the distribution of $X \oplus Y$?

(b) With notation as in sub-problem (a), is $X \oplus Y$ independent of X? Is $X \oplus Y$ independent of Y? Be sure to consider both the case $p = 1/2$ and the case $p \neq 1/2$.

(c) Let $X_1, ..., X_n$ be i.i.d. (i.e., independent and identically distributed) Bern(1/2) R.V.s. For each nonempty subset J of $\{1, 2, ..., n\}$, let

$$Y_J = \oplus_{j \in J} X_j.$$

Show that $Y_J \sim \text{Bern}(1/2)$ and that these $2^n - 1$ R.V.s are pairwise independent, but not independent.

(a)

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	0

$X \sim \text{Bern}(p)$. $P(X=1) = p$, $P(X=0) = 1-p$

$Y \sim \text{Bern}(\frac{1}{2})$, $P(Y=1) = P(Y=0) = \frac{1}{2}$.

$Z = X \oplus Y$

$P(Z=1) = P(X=0, Y=1) + P(X=1, Y=0) = \frac{1}{2}(1-p) + \frac{1}{2}p = \frac{1}{2}$.

$P(Z=0) = P(X=Y=0) + P(X=Y=1) = \frac{1}{2}p + \frac{1}{2}(1-p) = \frac{1}{2}$.

(b) if X is known. $Z = Y$ OR $Z = \bar{Y}$, $P(Z=0) = P(Z=1) = \frac{1}{2}$, do not change the probability,

so $X \oplus Y$ is independent of X

if Y is known. $Y=0$, $P(Z=1) = P(X=1) = p$. $Y=1$, $P(Z=1) = 1-p$.

if $p = \frac{1}{2}$, $X \oplus Y$ is independent of Y,

if $p \neq \frac{1}{2}$, $X \oplus Y$ is not independent of Y.

(c) $Y_J = \oplus_{j \in J} X_j$.

fix all the variables except X_j in Y_J. then Y_J = either X_j OR $1-X_j$, since $X_j \sim \text{Bern}(\frac{1}{2})$,

therefore $Y_J \sim \text{Bern}(\frac{1}{2})$

Let J and K be two distinct nonempty subsets, after interchanging the names, choose an index $j \in J/K$.

so fix all other variables except X_j, Y_K is fixed. Y_J = either X_j or $1-X_j$. $X_j \sim \text{Bern}(\frac{1}{2})$, $\Rightarrow Y_J \sim \text{Bern}(\frac{1}{2})$

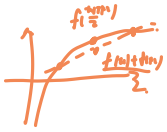
so knowing one of Y_K, Y_J do not change the probability, they are pairwise independent

but they are not mutually independent, think of X_1, X_2 and $X_3 = X_1 \oplus X_2$, then they are dependent.

Problem 4

Let X be a discrete R.V. whose distinct possible values are $a_1, a_2, ..., a_n$, with probabilities $p_1, p_2 ..., p_n$, respectively (so $p_1 + p_2 + \cdots + p_n = 1$). The entropy of X is defined to be the average surprise of learning the value of X:

$$H(X) = \sum_{j=1}^{n} p_j \log_2(1/p_j).$$



Show that the maximum entropy for X is when its distribution is uniform over $a_1, a_2, ..., a_n$, i.e., $p_j = 1/n, \forall j$.

assume $p_j > 0$, according to Jensen's inequality for $f(x) = \log_2 x$.

$$\sum_{i=1}^{n} a_i f(x_i) \le f\left(\sum_{i=1}^{n} a_i x_i\right)$$

so $H(X) = \sum_{j=1}^{n} p_j \log_2(1/p_j) \le \log_2\left(\sum_{j=1}^{n} p_j \cdot \frac{1}{p_j}\right) = \log_2 n.$

equality holds if and only if $p_j = 1/n$, $\forall j$.